



UNIVERSITY OF SCIENCE
HO CHI MINH CITY

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A Few Web and Mobile UI Patterns

Outline

- 
- Overview
 - Web UI Patterns
 - Navigation
 - Forms
 - Search
 - Wizard
 - Mobile UI Patterns
 - Navigation
 - Forms
 - Search
 - Notification, feedback, and invitation

What is a UI Design Pattern?

- Object-oriented Methodology
 - “Design patterns are descriptions of communicating objects and classes that are customized to solve a general design problem in a particular context” (Erich Gamma et al. 1995)
- Generally, a design pattern is a description of the repeatable problem and its solution
- UI design patterns are descriptions of common UI design problems and their solutions

Benefits of UI Design Patterns

- UI patterns support external consistency
 - Consistency is a key principle of UI design
- Improve learnability, memorability, and satisfaction of UI
- Reduce time to design, develop, and test
- Reduce risks in UI design
 - Learning from good and working UI's

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- Mobile UI Patterns

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Web Navigation

- Current popular homepage design: menus on top

Logo, menu, sub-menu, search input box

Sliders: conveying the current most important messages

Content

Fat footer

Navigation: Menus and Sliders

- Horizontal primary menu and submenu
- Menu can have dropdown lists (vertical menus)
- Sliders should include graphics and most important messages
- Sliders can automatically play, but should not play forever
 - Include pause and replay buttons
 - Don't play too fast, let users read the message
- Search input box on top should be prominent
- Limit scroll on homepage
- Always provide a Home link

Navigation: Menus and Sliders

Search box

The screenshot displays the USC University of Southern California website. At the top, a dark navigation bar contains links for students, alumni, parents, faculty, employees, and sports. To the right of these links is a 'popular links' section and a search box labeled 'search usc'. Below the navigation bar is a red banner featuring the USC logo and the text 'USC University of Southern California'. Underneath the banner is a horizontal menu with links for about, administration, admission, academics, research, patient care, community, arts, and global initiatives. On the right side of this menu, the text 'Los Angeles 58°' is displayed. The main content area features a large slider image of a conductor leading an orchestra. A red overlay on the left side of the slider contains the text 'Tilson Thomas comes home' and a paragraph about Michael Tilson Thomas. Below the slider image, there is a row of four small thumbnail images. At the bottom right of the page, the 'Los Angeles Times' logo is visible.

students alumni parents faculty employees sports

popular links search usc

USC University of Southern California

about administration admission academics research patient care community arts global initiatives

Los Angeles 58°

Tilson Thomas comes home

USC Thornton School of Music alumnus **Michael Tilson Thomas** has been named a Judge Widney Professor of Music at USC. He begins his appointment in fall 2015.

Photo: USC/Steve Cohn

Los Angeles Times

Multiple sliders
(don't animate
forever)

Navigation: Vertical Dropdown Menu

[students](#) [alumni](#) [parents](#) [faculty](#) [employees](#) [sports](#) [popular links](#) [search usc](#)

 **USC** University of Southern California

[about](#) [administration](#) [admission](#) [academics](#) [research](#) [patient care](#) [community](#) [arts](#) [global initiatives](#) Los Angeles 58

[Board of Trustees](#)

[President](#)

[Senior Administration](#)

[Academic Deans](#)

"To be able to lead this Trojan Family forward now is the opportunity of a lifetime," said USC's 11th president, C. L. Max Nikias, "as we write together the next chapter in USC's extraordinary history." Nikias shared his vision for USC's future during his inaugural address, "[The Destined Reign of Troy](#)."



An accomplished researcher, educator and university administrator, President Nikias has earned accolades for his leadership, innovation and fund raising as well as his ability to build partnerships among varied constituencies.



Tilson Thomas comes home

USC Thornton School of Music alumnus [Michael Tilson](#)

Fat Footer

- Secondary navigation to provide more shortcuts to important information of a website
- Links to frequently used pages
- Using the same footer for all pages

Other Microsoft sites  Windows  Office  Surface  Windows Phone  Xbox  Skype  Bing  Microsoft Store	Downloads Download Center Windows downloads Update to Windows 8.1 Free downloads Internet Explorer Office downloads Drivers (sound, printer, USB, others) Subscribe Microsoft At Home RSS Microsoft At Work RSS Microsoft newsletters	Support Support home Microsoft Security Essentials Microsoft Update: Service packs & fixes Windows help & how-to Office (Outlook, Excel, and others) About Microsoft Microsoft Account Careers Company news Investor relations Site map	Popular resources Microsoft Cloud PC and Laptops Tablet and Computers Windows Phone smartphones Windows Phone apps and games Microsoft computer security Malware removal tool Microsoft Dynamics CRM Windows Apps
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 [United States - English](#)



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Breadcrumbs

- Tell users where they have been, where they are, how they can go back

The screenshot displays the VMware vCloud Suite product page. At the top, the VMware logo is on the left, and navigation links for Products, Support, Downloads, Consulting, Partners, and Company are on the right. Below the logo, a breadcrumb trail is highlighted with a red box, showing the path: Home / Products / vCloud Suite. To the right of the breadcrumb, there is a link to 'Email Us' with the phone number 1-877-486-9273. Below the breadcrumb, the product name 'vCloud Suite' is displayed in a large blue font. Underneath the product name is a horizontal menu with tabs for Overview, Features, Resources, Compare, and Pricing. The 'Overview' tab is currently selected. Below the menu, a paragraph of text describes the product: 'VMware vCloud® Suite lets you build and run a vSphere-based private cloud that delivers cloud service provider economics at scale, application provisioning in minutes, and automated operations management.'

Pagination

- Using 'Previous', 'Next', '<' and '>' icons, and page numbers
- Google:



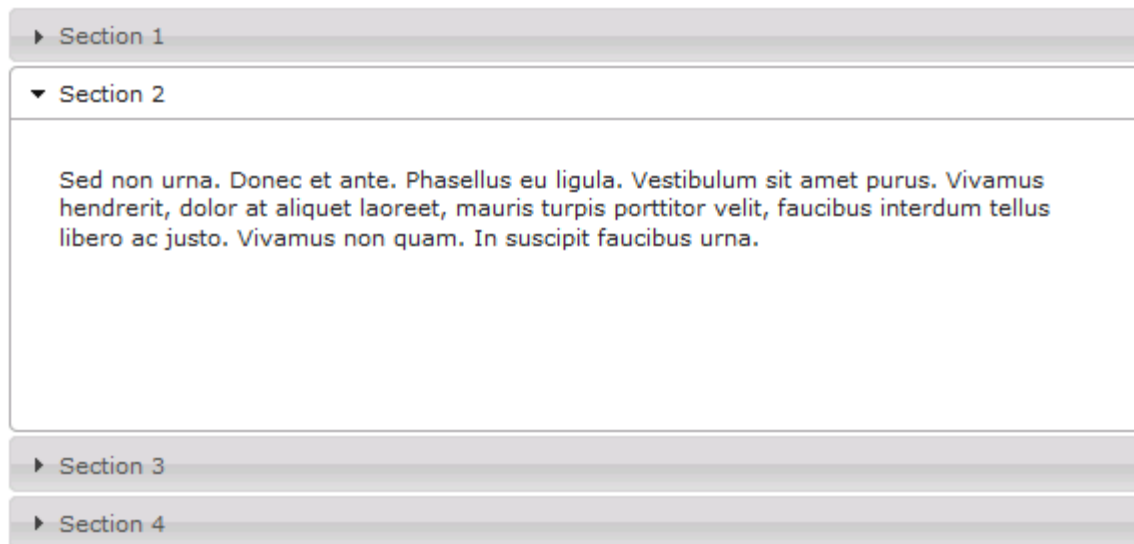
- ebay:



Allow users to change number of items per page:
Too many: slow loading
Too few: slow searching

Accordion Menu

- Grouping things into sections each with a panel
- Expanding a section when clicking on it
- Allowing users to view content of a section while seeing other sections



Source: <http://jqueryui.com/accordion/>

Carousel

- Used to browse through a set of items
- Used when there are many items and limited space while informing users of something more
- Used for visual objects

Continue Shopping: Customers Who Bought Items in Your Recent History Also Bought

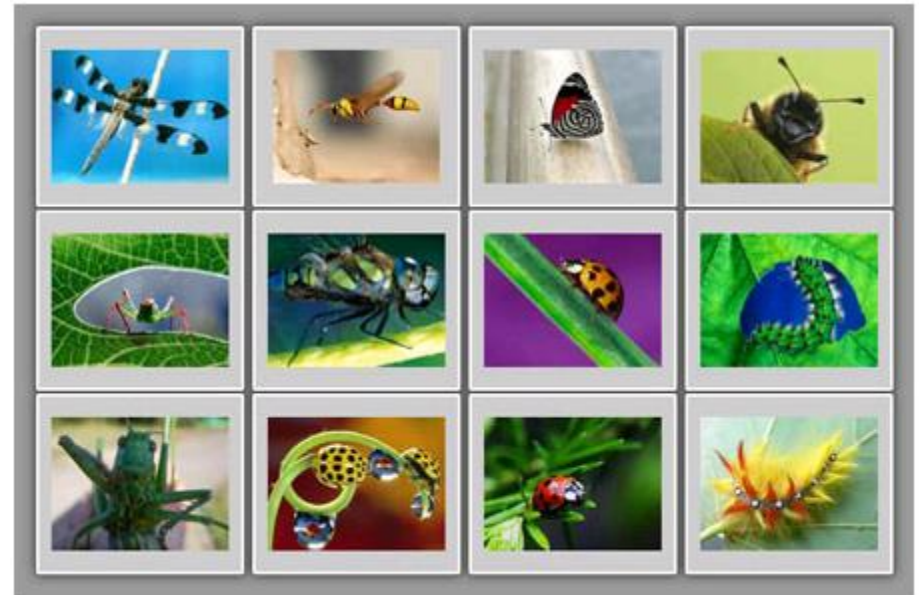
Page 12 of 13 [Start over](#)



Thumbnail Gallery

- Allowing users to view many images without downloading full sized content
- Many jQuery and dojo plugins available
- Common sizes:

Width x Height	Description
64×64	Small
80x80	Medium
96x96	Medium
128x128	Large
144x144	Extra large



Source: <http://www.dmxzone.com/go/17142/creating-a-simple-animated-thumbnail-gallery/>

Forms

- Forms contain various input elements: text boxes, combo boxes (dropdown), radio buttons, etc.
- Guidelines
 - ❑ Use one-column left aligned forms
 - ❑ Mark required fields or active feedback on required fields
 - ❑ Provide good feedback for actions and good error messages
 - ❑ Place labels on the left or top and consistent throughout the website
 - ❑ Group related fields (using colors or shading)

Forms (cont'd)

- Input prompts placed within input box
- Good defaults
- Good feedback
- Few required info

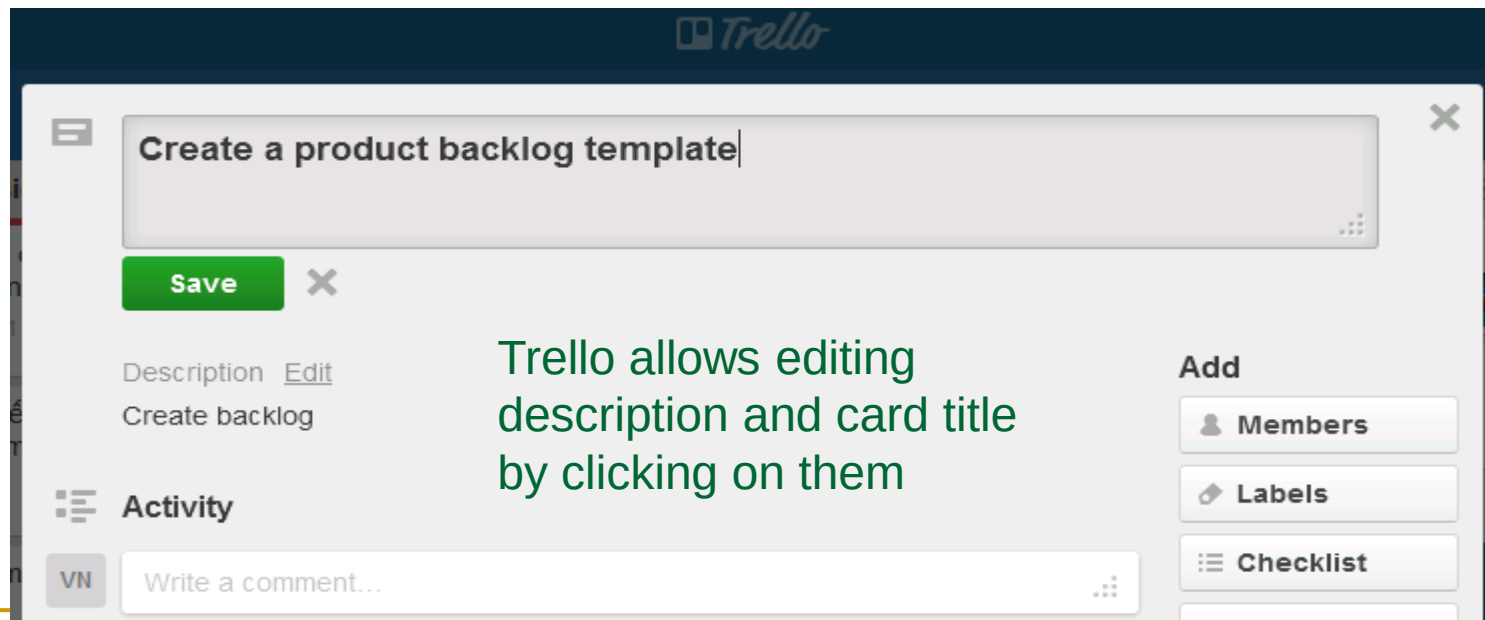
Required fields indicated when users leave them without filling

This form does not use many labels: saves space and time for users

The screenshot shows the Yahoo! 'Sign up' form. At the top, the 'YAHOO!' logo is on the left, and 'Sign In' and 'Help' links are on the right. Below the logo, the text 'English (United States)' is displayed. The main heading is 'Sign up'. The form contains several input fields: 'First name' and 'Last name' (both with red borders and red error text 'Enter first name.' below them), 'Yahoo username' (with a red border and red error text 'This information is required.' below it, and a default value '@yahoo.com' on the right), 'Password' (with a red border and red error text 'This information is required.' below it, and a 'show' checkbox on the right), 'Mobile number' (with a red border and red error text 'This information is required.' below it, and a dropdown menu showing '+1' and a country code icon), 'Birthday' (with a red border and red error text 'Your full birthday is required.' below it, and a dropdown menu showing 'Month', 'Day', and 'Year'), and 'Optional recovery number' (with a red border and red error text 'This information is required.' below it, and a dropdown menu showing '+1' and a country code icon). There are also radio buttons for 'Male' and 'Female' (with 'Female' selected), and a 'Relationship' dropdown menu. The form is annotated with green lines pointing to the red error text for the 'First name', 'Yahoo username', and 'Mobile number' fields.

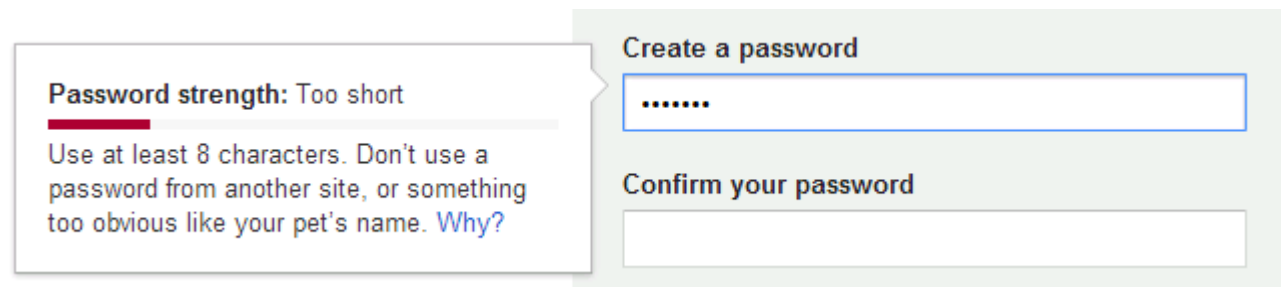
Inplace Editor

- Allows users to edit value easily
- Used for editing a few text, dropdown, etc. values
- Should use signifiers or indicators for possible inplace editors (e.g., highlight when hovering)



Password Strength Meter

- Used in the sign up form to help users specify their strong passwords
- Helps improve password security on websites
- How to know a password is strong?
 - Apply some simple policies, e.g., [wiki page](#) or [MS page](#)

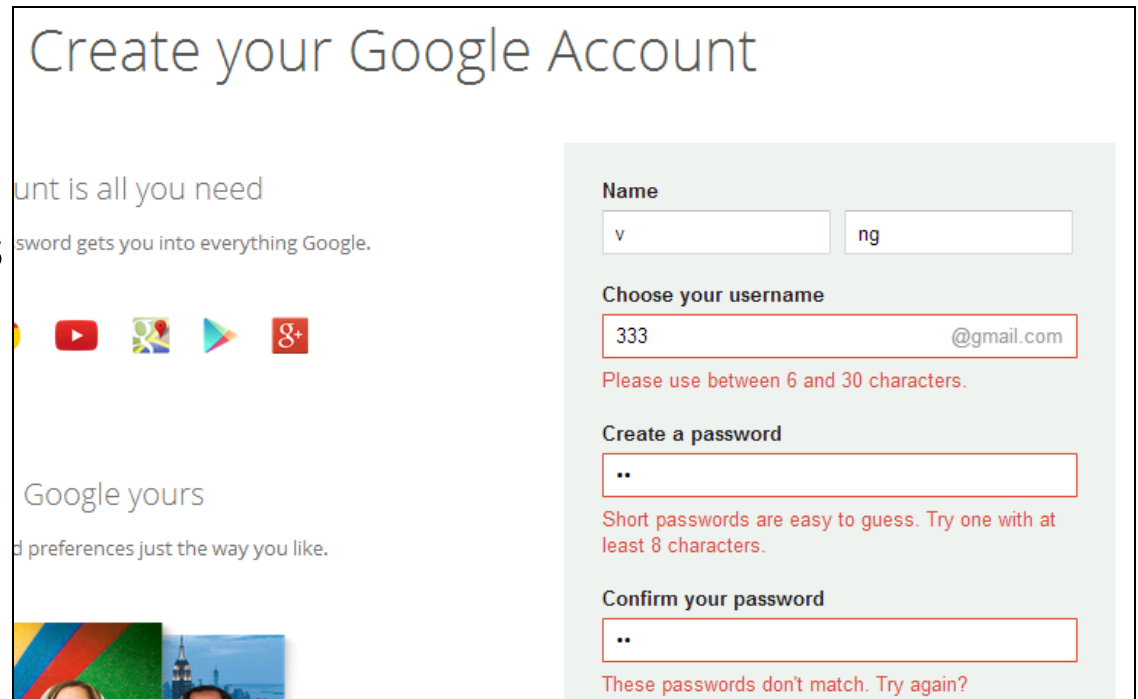


The image shows a portion of a web form for creating a password. On the left, a white box with a purple border contains a warning: "Password strength: Too short" followed by a red progress bar. Below this, it says "Use at least 8 characters. Don't use a password from another site, or something too obvious like your pet's name. [Why?](#)". To the right, the form has two input fields. The first is labeled "Create a password" and contains six dots. The second is labeled "Confirm your password" and is empty.

Gmail's sign up form

Input Validation

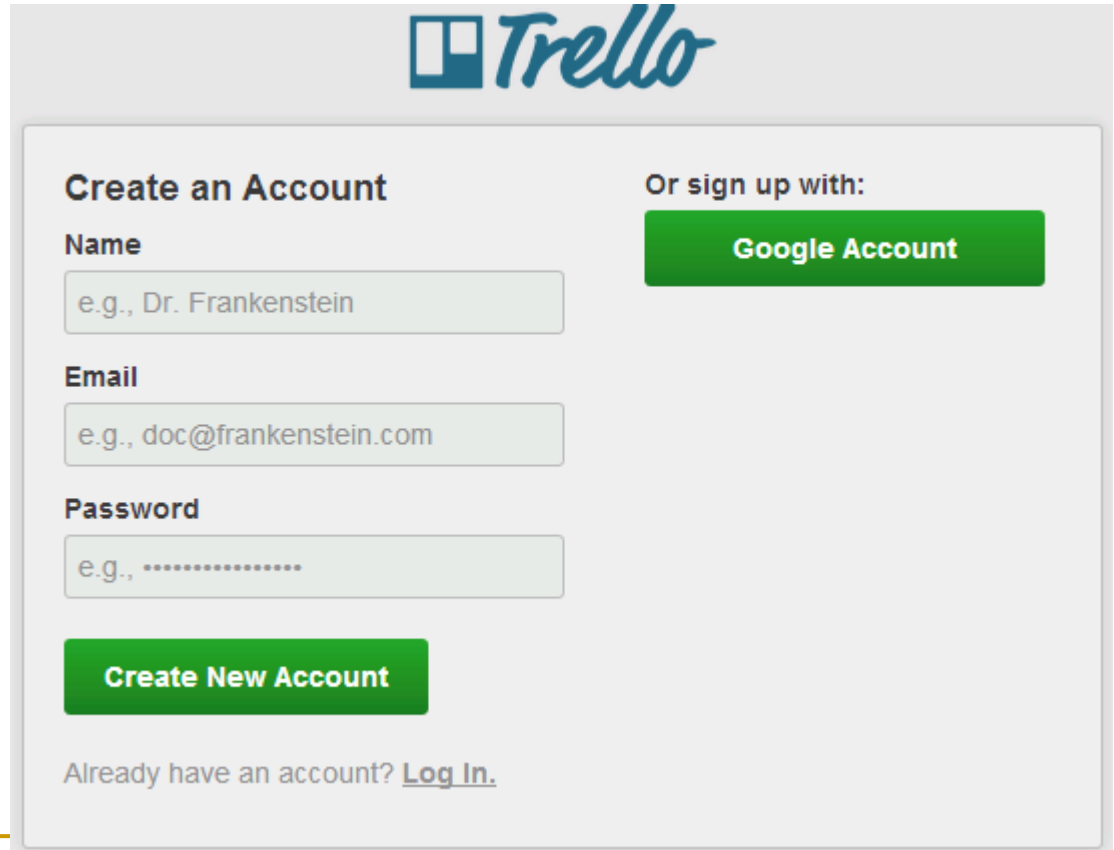
- Provide validation and feedback input time
- Validate and feedback as much as possible before submitting
- What to validate?
 - ❑ Required fields
 - ❑ Acceptable ranges
 - ❑ Length
 - ❑ Uniqueness
 - ❑ Confirmation
 - ❑ Inclusion
 - ❑ Exclusion



The screenshot shows the 'Create your Google Account' form. It includes fields for Name (first and last), Username, and Password. The Username field contains '333' and the email part is '@gmail.com'. The Password field contains '..'. Red error messages are displayed below the Username and Password fields: 'Please use between 6 and 30 characters.' for the username and 'Short passwords are easy to guess. Try one with at least 8 characters.' for the password. A confirmation field for the password is also present, containing '..', with a red message below it: 'These passwords don't match. Try again?'. The form also features social media icons (YouTube, Google+, etc.) and a section for 'Google yours' with a photo upload area.

Lazy Registration

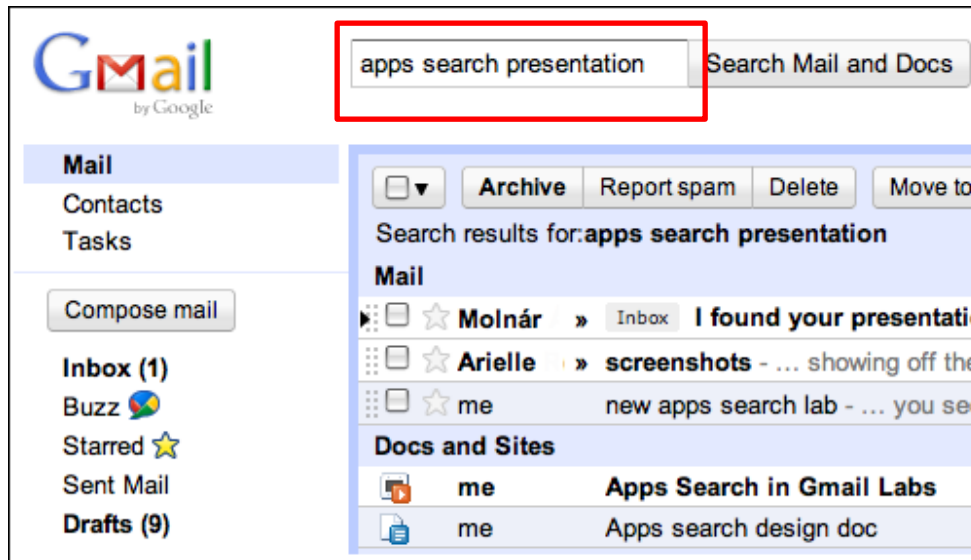
- Allows users to quickly register your service
- Why? Users are afraid of having a relationship with a website
- How to get more of their info?
 - Use “carrot and stick” approach



The image shows the Trello account creation interface. At the top is the Trello logo. Below it, the form is divided into two main sections. On the left, under the heading 'Create an Account', there are three input fields: 'Name' (with placeholder text 'e.g., Dr. Frankenstein'), 'Email' (with placeholder text 'e.g., doc@frankenstein.com'), and 'Password' (with placeholder text 'e.g.,'). Below these fields is a green button labeled 'Create New Account'. On the right, under the heading 'Or sign up with:', there is a green button labeled 'Google Account'. At the bottom of the form, there is a link that says 'Already have an account? [Log In.](#)'

Search

- Provide a single search field
 - Provide advanced search as a secondary option
- Show results in the same page with search criteria
- Offer an option to clear search criteria
- Offer a progress indicator during search



Search (cont'd)

- Search criteria is also shown with results

www.orbitz.com/shop/home?type=hotel&hotel.type=keyword&hotel.coord=&hotel.keyword.key=San+Francisco%2C+CA%2C+United+States&ho

ORBITZ Hotels Flights Vacation Packages Cars Cruises Deals Activities **New! ORBITZ REWARDS** Welcome

EXCLUSIVE SAVINGS Get an extra 15% off our best hotel deals Use promo code: **BYEBYE15** Learn more

Change search

Where
San Francisco, CA, United :

Check-in 4/15/14 **Check-out** 4/16/14

Nights: 1

Rooms 1 **Guests** 2

Search

[View hotels on map](#)

[Reset filters](#)

☐ Hotel name

301 matching hotels found in San Francisco (and vicinity) Previous | [Next](#)

Sort by: Best Bets [Lowest Price](#) [Distance](#) [Star Rating](#) [Reviewer Score](#)

The Opal San Francisco
★★★★★ 3.5 / 5 [569 reviews](#)
↓ Price is 19% less than usual
Union Square-Nob Hill
Historic hotel with spacious suites near Union Square
ORBITZ REWARDS [Join/sign in now](#) to earn.
There are 4 people viewing this hotel.
Last booked 38 minutes ago.

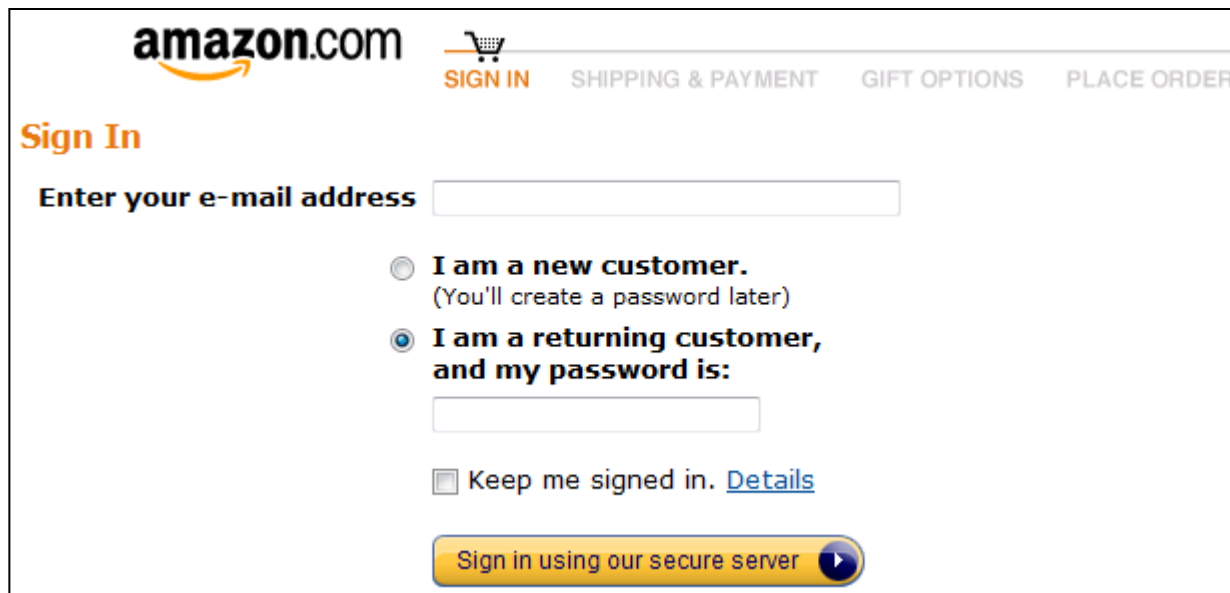
\$118
Avg/night
FREE Cancellation
See details
Earn \$4.14 in Orbucks

Hotel Rex, a Joie de Vivre Hotel
★★★★★ 4.2 / 5 [70 reviews](#)
↓ Price is 43% less than usual
Union Square-Nob Hill
Literary inspired hotel in historic San Francisco
ORBITZ REWARDS [Join/sign in now](#) to earn.
Last booked 12 hours ago.

\$157
Avg/night
FREE Cancellation
See details
Earn \$5.48 in Orbucks

Progress Indicator/Wizard

- Used to guide users through a complex multiple step process such as checkout, setup, installation
 - Also used when users are not familiar with infrequent processes



The screenshot displays the Amazon.com Sign In page. At the top, the Amazon logo is on the left, and a shopping cart icon is on the right. Below the cart icon, a progress indicator shows four steps: 'SIGN IN' (highlighted in orange), 'SHIPPING & PAYMENT', 'GIFT OPTIONS', and 'PLACE ORDER'. The main heading 'Sign In' is in orange. Below it, the text 'Enter your e-mail address' is followed by a text input field. Two radio buttons are present: 'I am a new customer. (You'll create a password later)' and 'I am a returning customer, and my password is:'. The second option is selected. Below the radio buttons is a password input field. A checkbox labeled 'Keep me signed in. Details' is also visible. At the bottom, there is a yellow button with the text 'Sign in using our secure server' and a play icon.

Progress Indicator/Wizard (cont'd)

■ Guidelines

- ❑ Minimize number of steps
- ❑ If more than 5 steps, group them and don't use numbers
- ❑ Show the current step clearly
- ❑ Label all steps/groups clearly
- ❑ Allow users to go back to modify inputs in previous steps

Outline

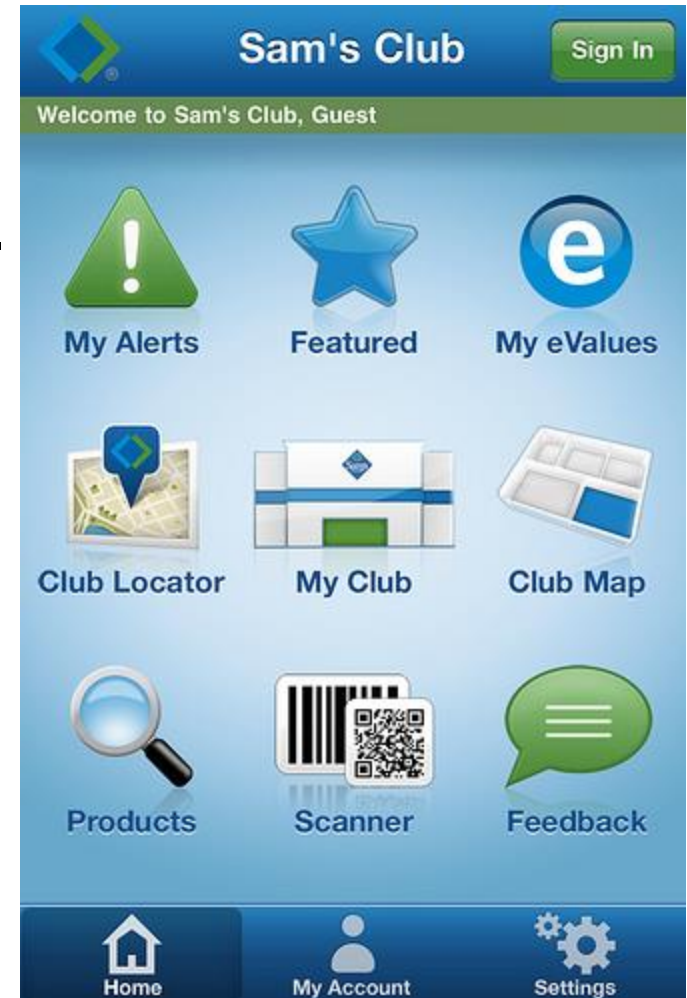
- Overview
- Web UI Patterns
 - Navigation
 - Forms
 - Search
 - Wizard
-  ■ Mobile UI Patterns
 - Navigation
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 - Search
 - Notification, feedback, and invitation

Mobile Navigation

- Types (Neil 2012)
 - Springboard
 - Tab menu
 - List menu
 - Gallery
 - Metaphor

Springboard

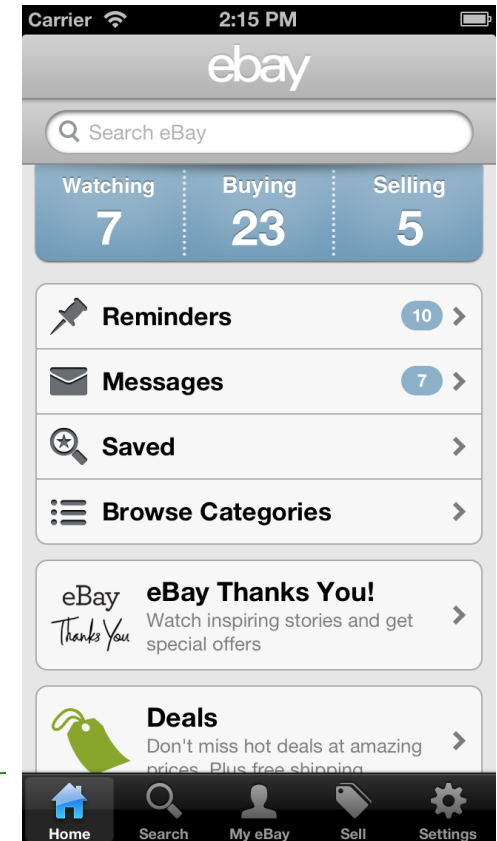
- Home screens for iOS and Android
- Divide screen into 3x3, 2x3, etc.
- Should limit number of items in one screen
 - Combine w/ tab if needed more
- Provide a search box to search items
- Allow users to change layout



Tab menu

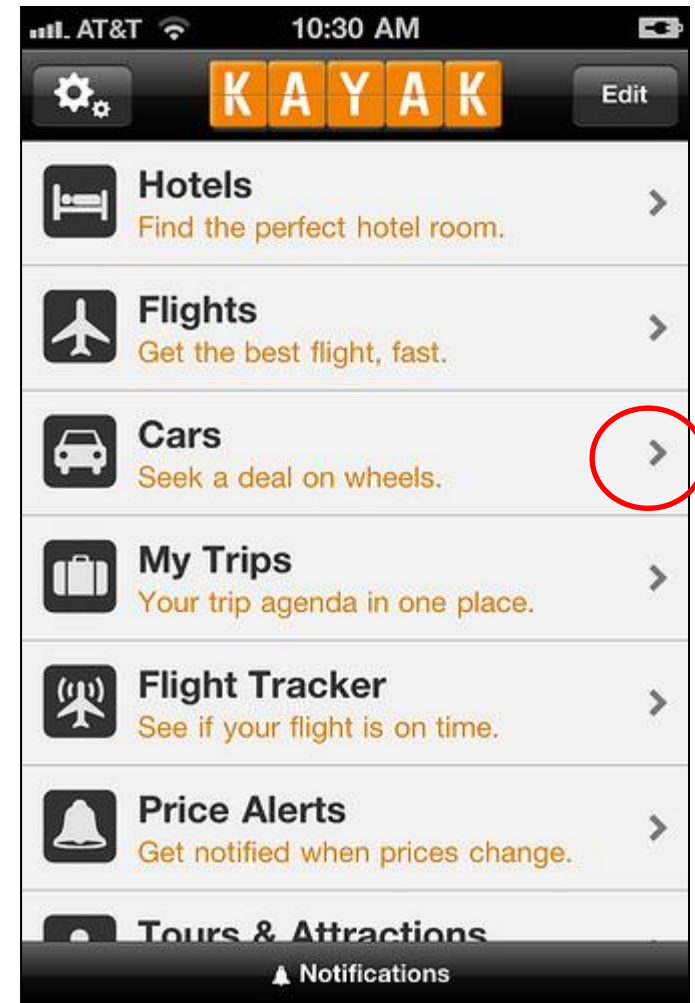
- Each tab groups a similar set of functions
- Active tab should be highlighted
- Tabs at the bottom is preferable (closer to user fingers)
 - Android, Windows: tabs on top
- Use icons with accompanying titles
- Titles should be short (use list menu or springboard for long titles instead)

tabs



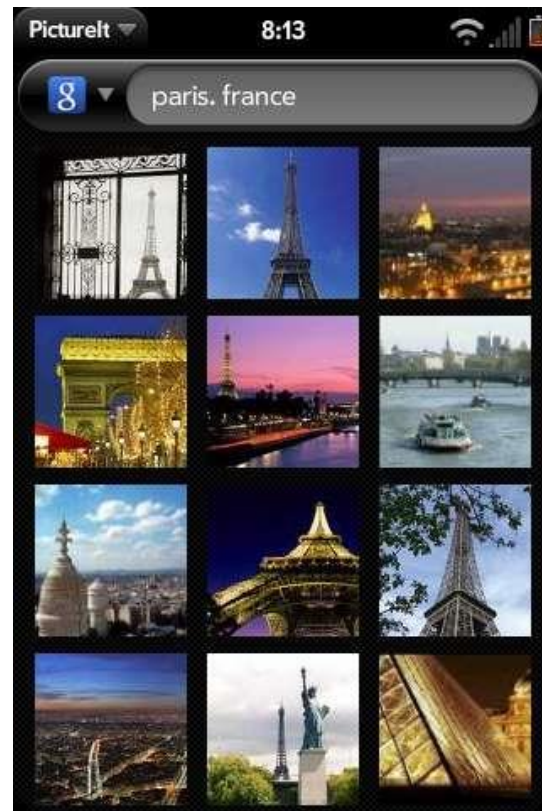
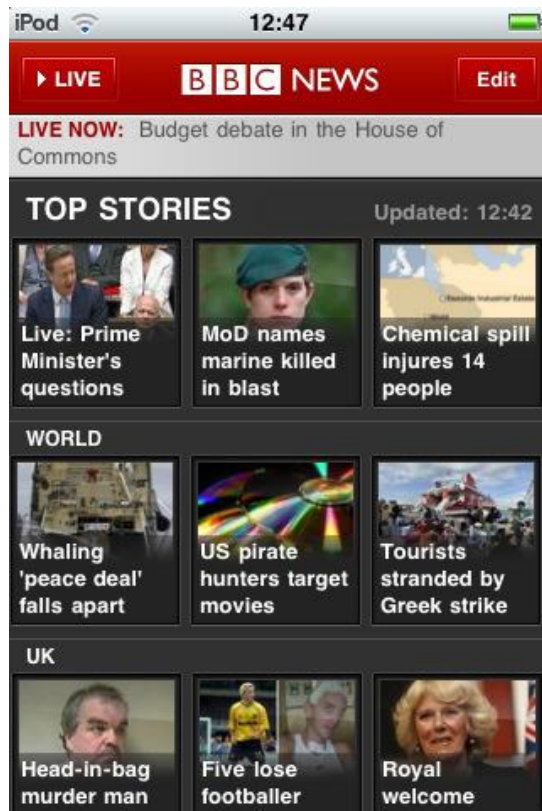
List Menu

- Various kinds of list
 - Title only list
 - Grouped list (with sub-sections)
 - Enhanced list
- Can show long titles and sub-titles (e.g., hints)
- Can be included in tabs
- Should indicate touchable items



Gallery

- Suitable for photos, articles, products, restaurant menus



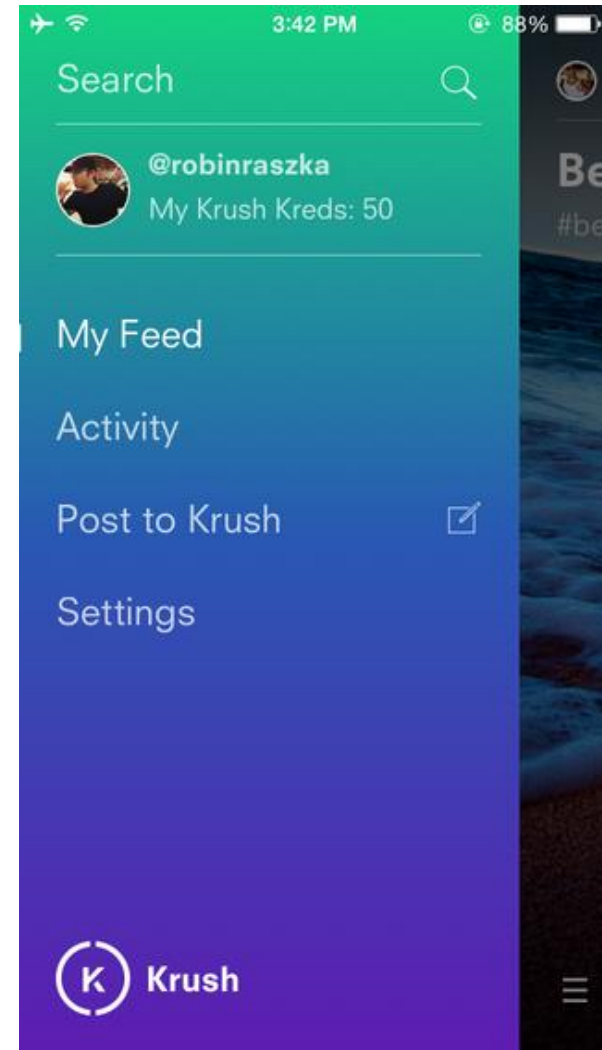
Metaphor

- Menu with icons symbolizing physical things
- Strong metaphors may be overkilled
- May be suitable in games



iOS 7 – Slide Between Pages

- Moving screens by sliding
- Less visible navigation but efficient



Forms

- Keep forms simple
- Only include required fields

iPod 6:55 PM

Cancel Create Account Done

Full Name John Appleseed

Skype Name Required

Password Required

Repeat Required

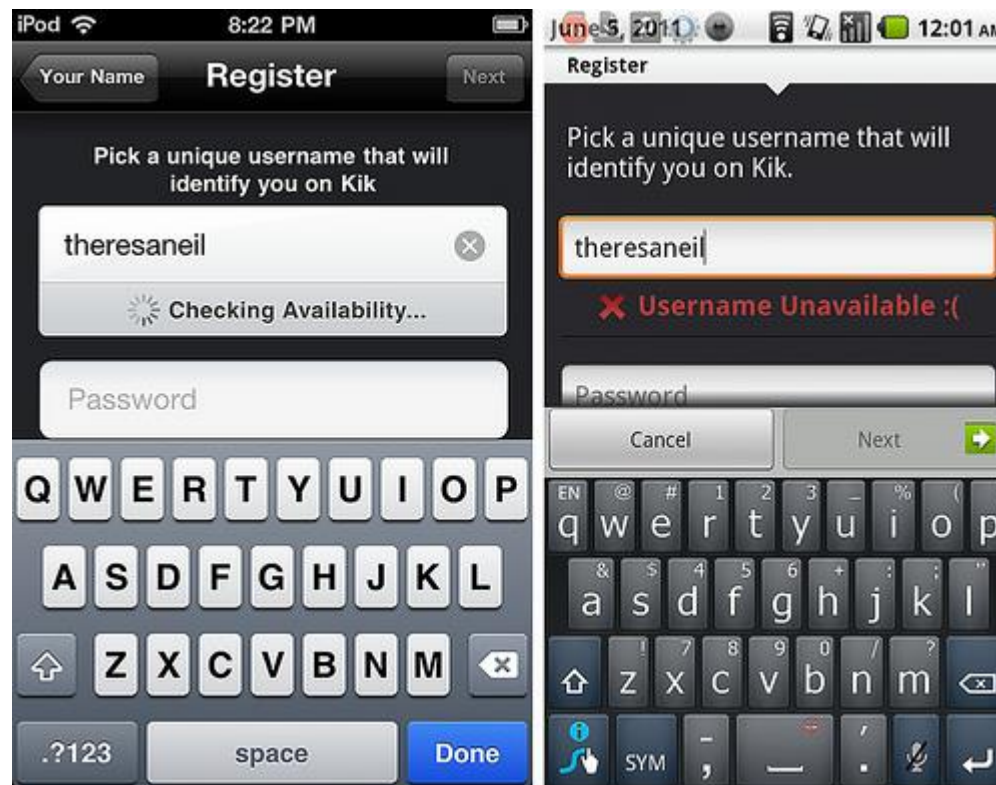
Email john@example.com

You must enter a valid email address in case you need to recover a lost password.

Get news and offers ON

Registration Form

- Provide instant feedback



Search Criteria Form

- Show criteria in a list
- Provide default values
- Have a “clear”/ “reset” button
- Set label with number of results, if possible

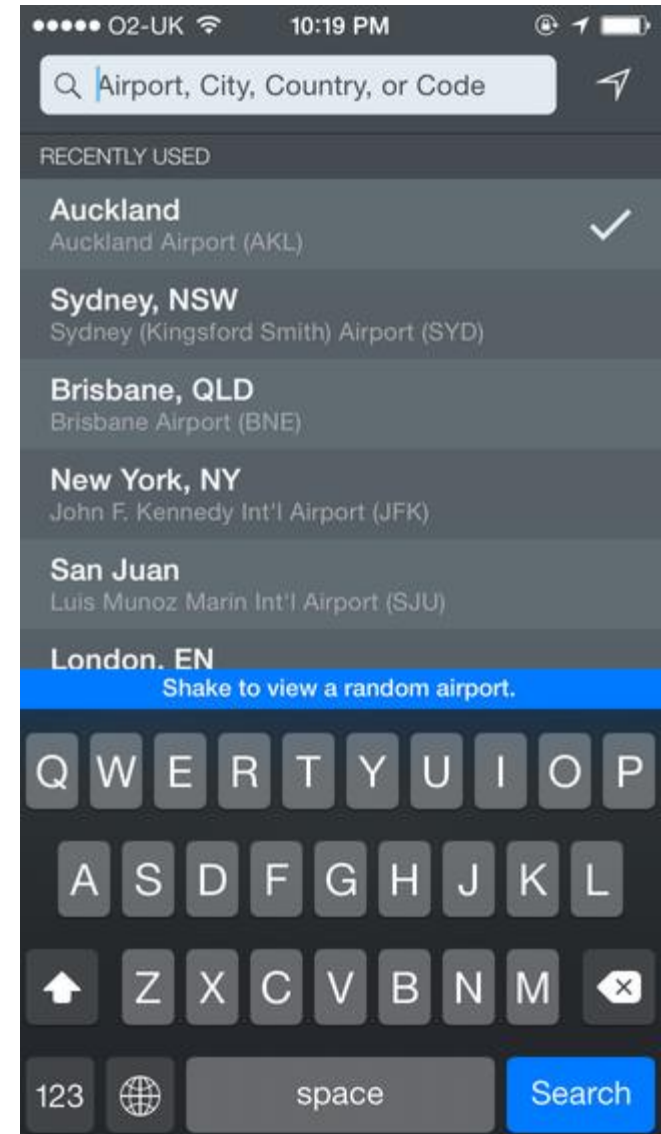
The screenshot shows the Edmunds New Car Inventory app interface on an iPod. The status bar at the top indicates 'iPod', signal strength, '5:36 PM', and battery level. The app's header is green with the Edmunds logo and the title 'New Car Inventory'. Below the header, there is a list of search criteria, each with a default value and a right arrow indicating a dropdown menu:

- Make/Model: MINI
- Year: Any
- Exterior Color: Select Model
- Engine Type: Any
- Price Range: Any
- Location: 78619 / 350 miles

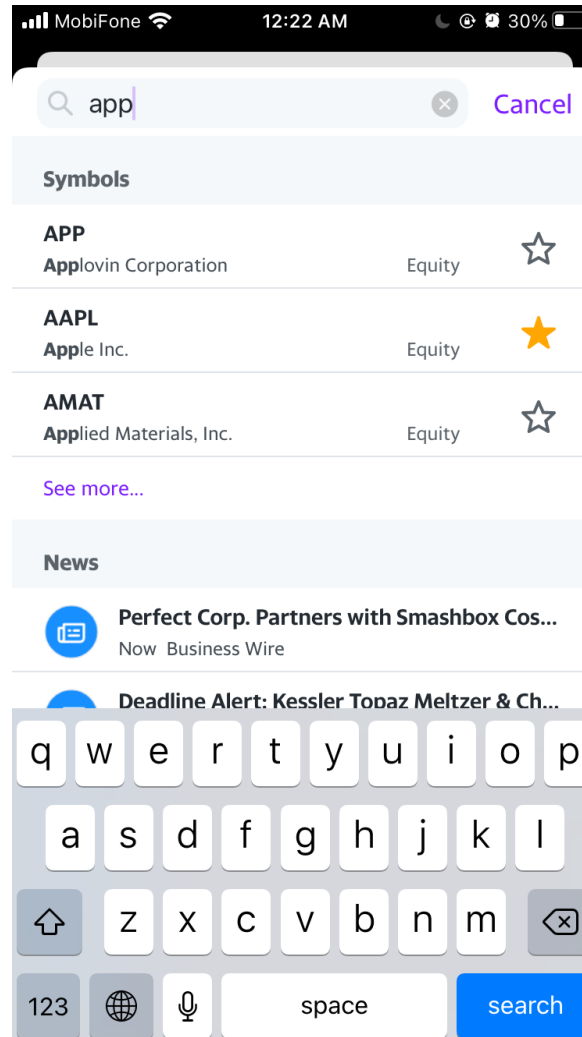
At the bottom of the search criteria list, there are two buttons: a large green button labeled 'Show 544 Matches' and a smaller grey button labeled 'Reset'. The bottom of the screen features a navigation bar with three icons: a dollar sign for 'Research', a car icon for 'Inventory' (which is highlighted), and a heart icon for 'Favorites'.

Single Search Input Box

- Always better to provide simple but powerful search capability



Search Auto Suggestion



Grouped Rows



iPod 12:10 PM

Accounts Free Checking

Mar 17, 2011

Home Depot	Home Improvement	-\$12.39	>
HEB	Groceries	-\$57.82	>
Freebirds	Restaurants	-\$18.39	>

Mar 16, 2011

Ruby Bbq	Restaurants	-\$37.72	>
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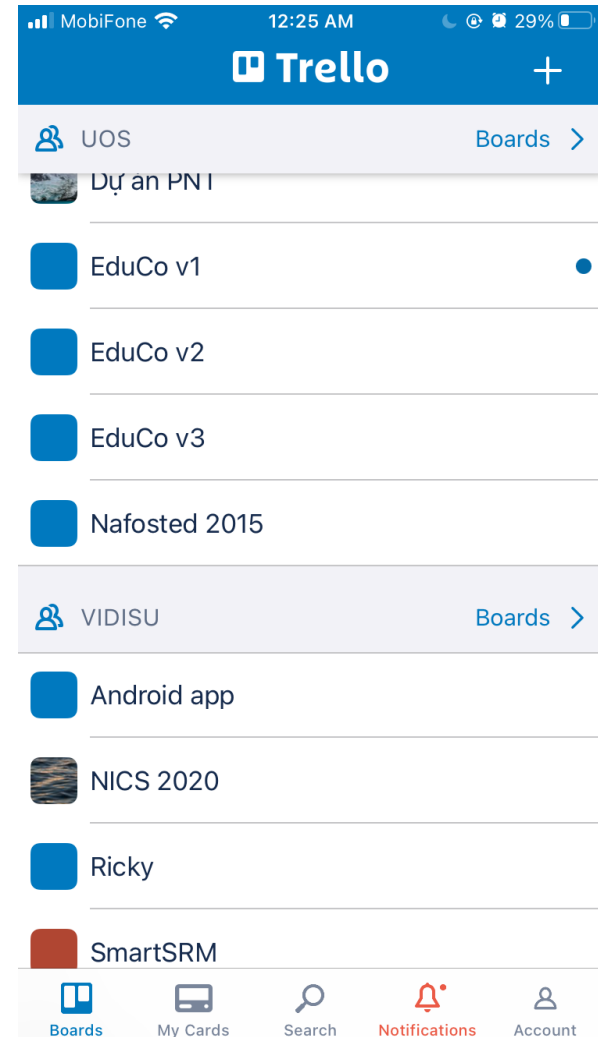
Mar 15, 2011

Check 1470	Check	-\$18.25	>
------------	-------	----------	---

Mar 14, 2011

ATM Fee		-\$3.00	>
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Updated 03/19/2011 at 12:10 PM



MobiFone 12:25 AM 29%

Trello

UOS Boards >

Dự án PN I

EduCo v1	
EduCo v2	
EduCo v3	
Nafosted 2015	

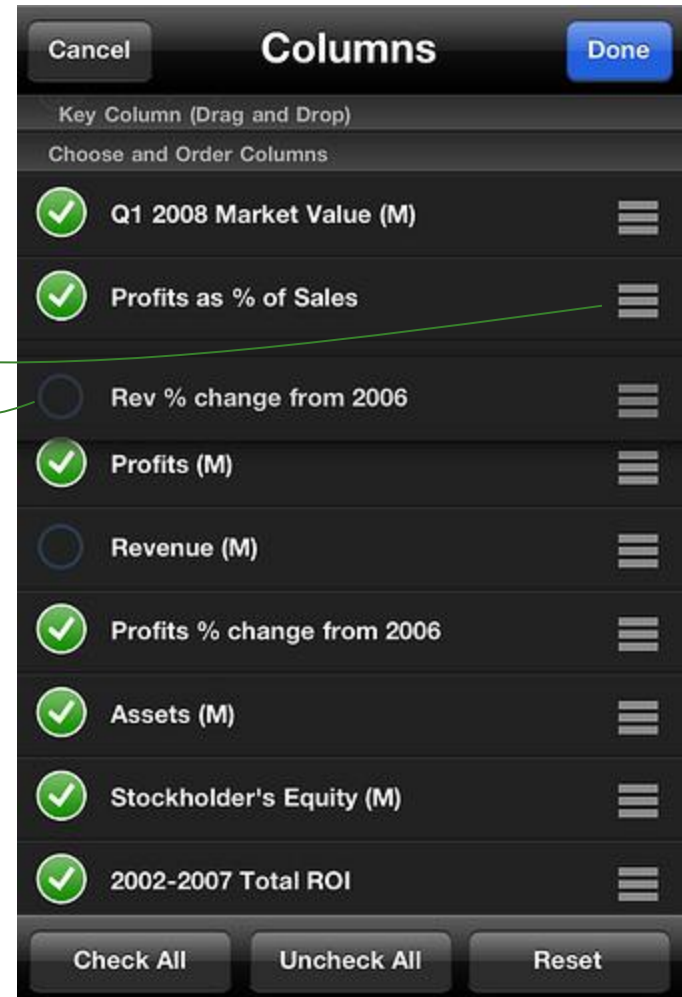
VIDISU Boards >

Android app	
NICS 2020	
Ricky	
SmartSRM	

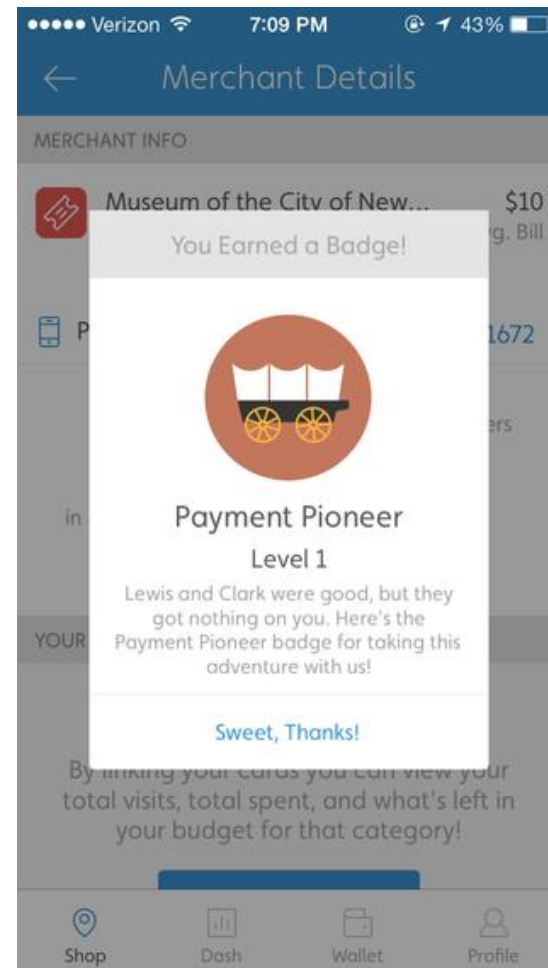
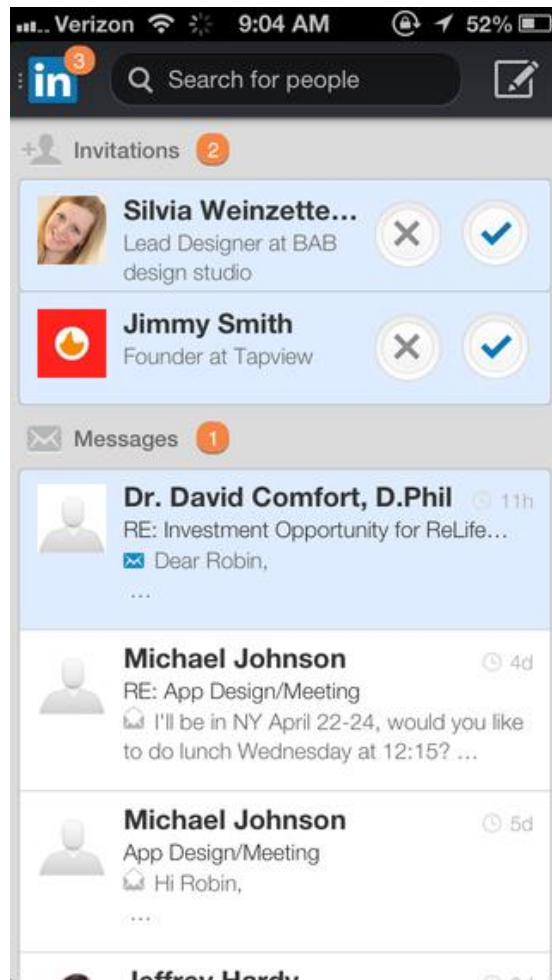
Boards My Cards Search Notifications Account

Signifiers

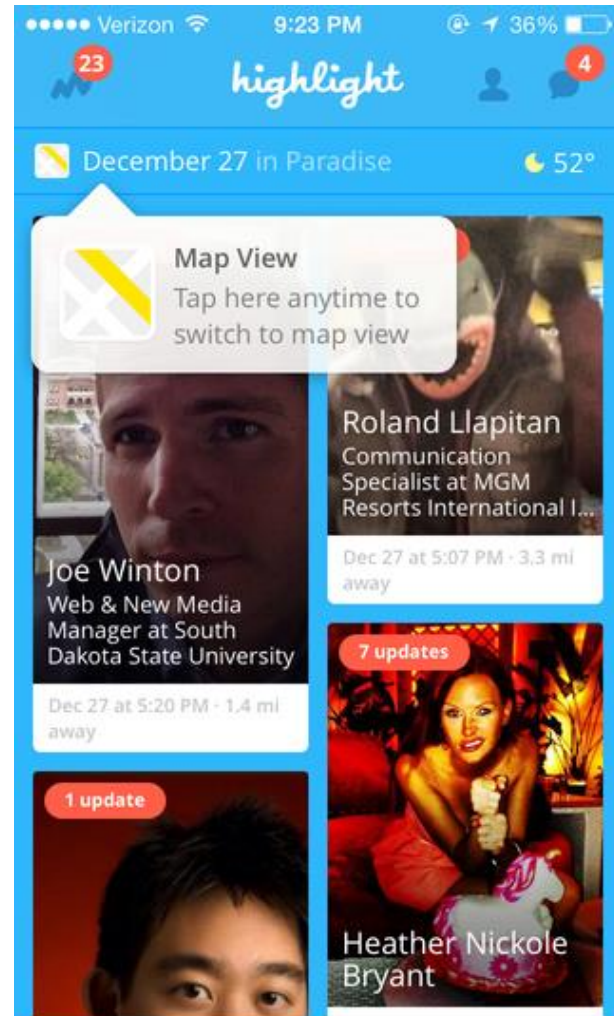
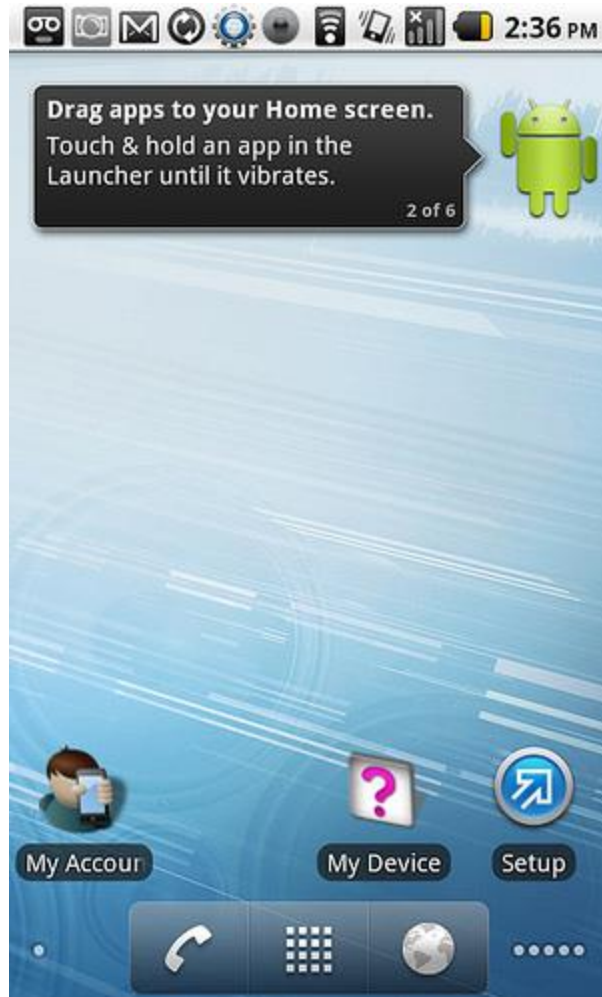
- Use signifiers/indicators to show affordance of items
- Drag-able
- Checkable



Notifications



Invitations



References

- iOS 7 UI Design Basics:
<https://developer.apple.com/library/ios/documentation/userexperience/conceptual/MobileHIG/index.html>
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<http://www.designingwebinterfaces.com/designing-web-interfaces-12-screen-patterns>
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